

INTERNSHIP MINOR PROJECT – CYBER SECURITY

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Internship Details:

Company Name: InternsElite
Program : Cyber Security Program
Duration : March 2026 – May 2026
Mentor : Mr. Aman Guta

Submitted To:

InternsElite

Date:

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CERTIFICATE

This is to certify that **Anil Kumar (Roll No: CS-T9/March-9349)** has successfully completed the **Minor Project in Cyber Security** as part of the Internship Program conducted by **InternsElite** during March 2026 – May 2026.

The work submitted is original and carried out under the guidance of the assigned mentor.

Mentor Signature

Mr. Aman Guta

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to **InternsElite** for providing me the opportunity to complete this Cyber Security Minor Project.

I am also thankful to my mentor **Mr. Aman Guta** for his continuous support, guidance, and encouragement throughout this project.

This project helped me gain practical knowledge of cybersecurity tools and techniques.

ABSTRACT

Cyber Security is a critical field that protects systems, networks, and data from unauthorized access and attacks. This minor project focuses on understanding various cyber threats and security tools used in ethical hacking.

The project includes study and practical understanding of network scanning, reconnaissance, malware types, password attacks, sniffing, and penetration testing tools.

All experiments were performed in a **controlled lab environment** for educational purposes only.

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INTRODUCTION

In today's digital world, cyber threats are increasing rapidly. Organizations face risks such as hacking, data theft, and malware attacks.

Cyber Security helps in protecting systems and ensuring data confidentiality, integrity, and availability (CIA triad).

This project focuses on basic cybersecurity concepts and tools used in ethical hacking and penetration testing.

1 . Explain the various types of Hacker/Hacking. And Consequences of Hacking in the Corporate sector.

Types of Hacker

- **White Hat Hacker** — Ethical security expert who protects systems legally.
- **Black Hat Hacker** — Malicious attacker who exploits systems for personal gain.
- **Grey Hat Hacker** — Mix of ethical and unethical hacking without permission.
- **Script Kiddie** — Beginner hacker using pre-made tools without deep knowledge.
- **Hacktivist** — Hacker driven by political or social causes.
- **State-Sponsored Hacker** — Government-backed hacker for cyber warfare and espionage.
- **Cyber Terrorist Hacker** — Hacker who uses cyber attacks to spread fear, disrupt systems, and achieve political or ideological goals.

Types of Hackers



White Hat Hackers

Ethical hackers improve security systems.



Black Hat Hackers

Malicious hackers exploit systems for gain.



Gray Hat Hackers

Blend of both Black hat & White hat Activities.



Script Kiddies

Inexperienced use pre-written tools.



Hacktivism

Promote political or social messages.



State-Sponsored Hacker

Conduct cyber espionage



Cyber Terrorists

Create fear and disruption for political/ideological reasons.

Types of Hacking

1. Network Hacking

- Target: Networks, servers
- Example: Sniffing, DoS attacks

2. Web Application Hacking

- Target: Websites & apps
- Example: SQL Injection, XSS

3. System Hacking

- Target: Operating systems
- Example: Password cracking, privilege escalation

4. Social Engineering

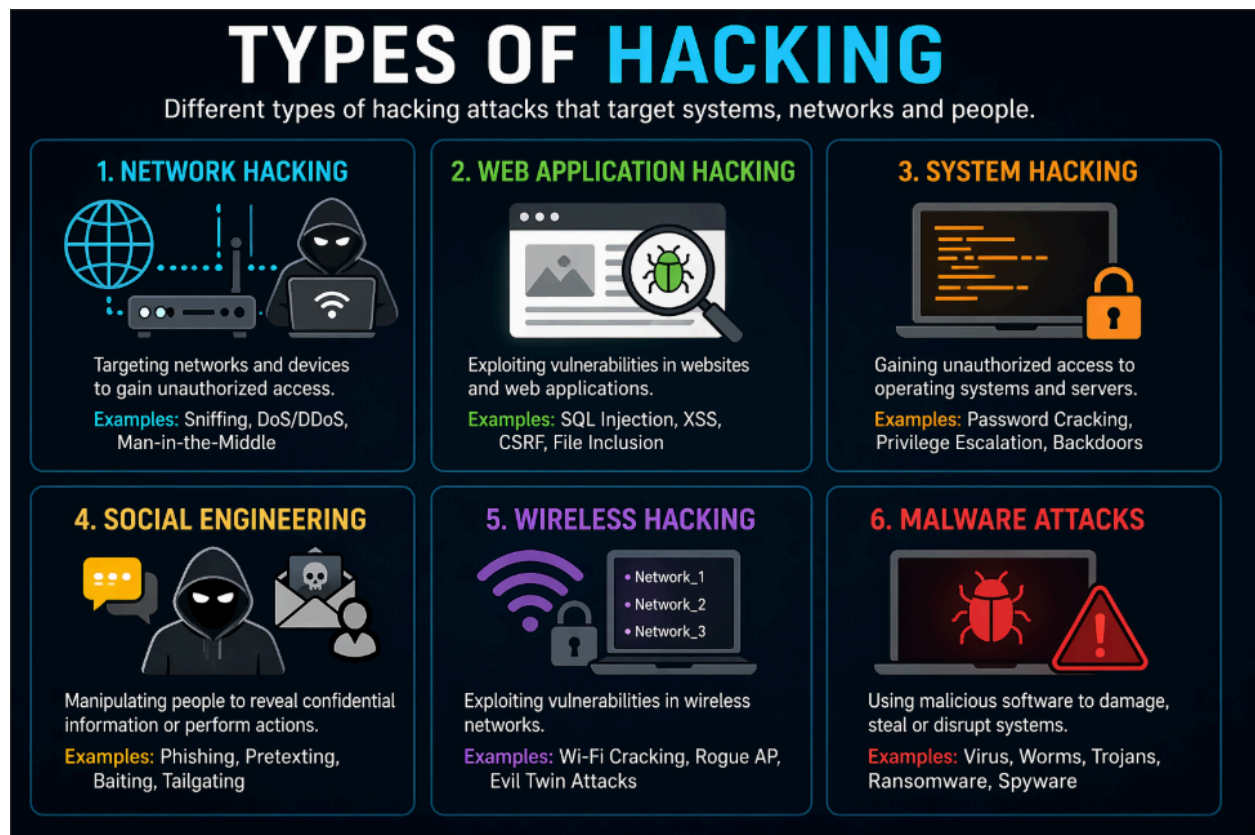
- Target: Humans
- Example: Phishing, pretexting

5. Wireless Hacking

- Target: Wi-Fi networks
- Example: WPA cracking, rogue access points

6. Malware Attacks

- Target: Devices via malicious software
- Example: Ransomware, Trojans



Consequences of Hacking in Corporate Sector

Hacking can severely impact organizations:

1. Financial Loss

- Theft of funds
- Cost of recovery, legal fees, fines

2. Data Breach

- Exposure of customer data, intellectual property
- Compliance violations (GDPR, HIPAA)

3. Reputation Damage

- Loss of customer trust
- Negative media coverage

4. Operational Disruption

- Systems downtime
- Business interruption (e.g., ransomware)

5. Legal Consequences

- Lawsuits from customers/partners
- Regulatory penalties

6. Intellectual Property Theft

- Loss of trade secrets
- Competitive disadvantage

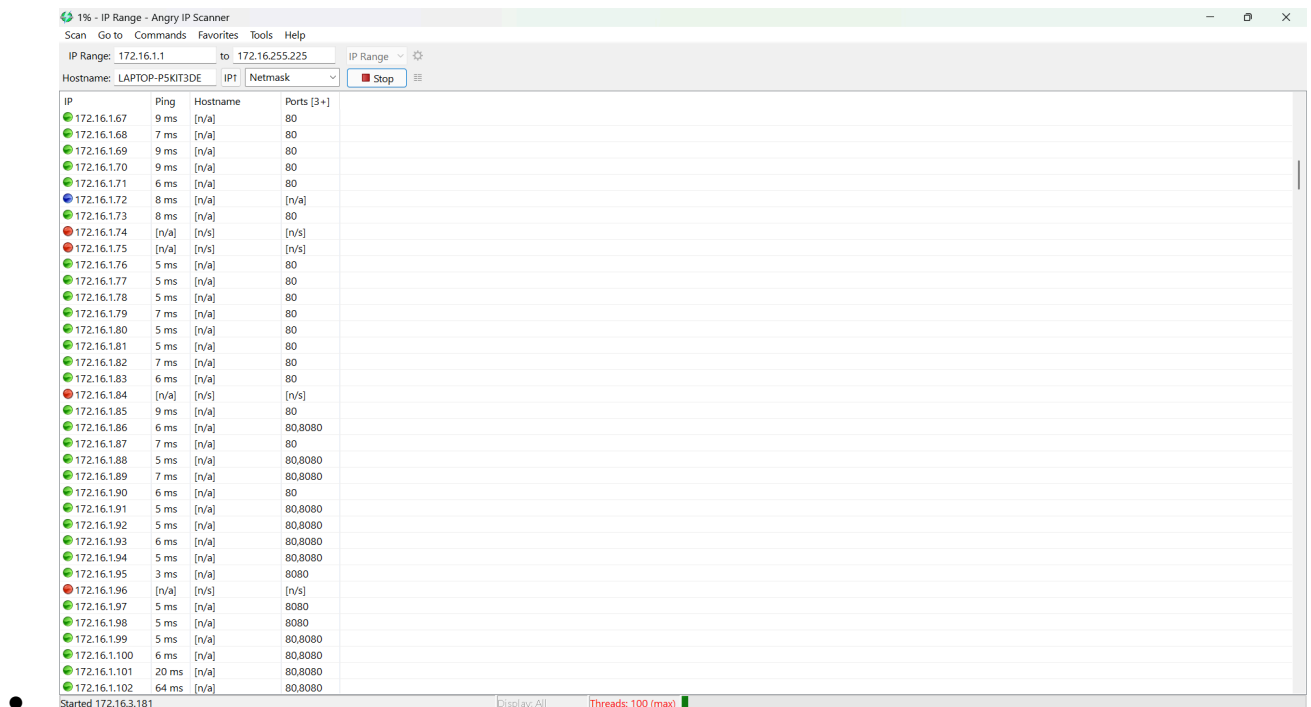
7. Employee Impact

- Job loss, stress, internal distrust

2. Explain the below terms. And perform its practical's.

1 . Scanning via Angry IP Scanner

- Tool used to **scan IP ranges**
- Finds **active hosts and open ports**
- Fast and simple network scanning



1% - IP Range - Angry IP Scanner

Scan Go to Commands Favorites Tools Help

IP Range: 172.16.1.1 to 172.16.255.225 IP Range

Hostname: LAPTOP-P5KIT3DE IP1 Netmask Stop

IP	Ping	Hostname	Ports [3+]
172.16.1.67	9 ms	[n/a]	80
172.16.1.68	7 ms	[n/a]	80
172.16.1.69	9 ms	[n/a]	80
172.16.1.70	9 ms	[n/a]	80
172.16.1.71	6 ms	[n/a]	80
172.16.1.72	8 ms	[n/a]	[n/a]
172.16.1.73	8 ms	[n/a]	80
172.16.1.74	[n/a]	[n/s]	[n/s]
172.16.1.75	[n/a]	[n/s]	[n/s]
172.16.1.76	5 ms	[n/a]	80
172.16.1.77	5 ms	[n/a]	80
172.16.1.78	5 ms	[n/a]	80
172.16.1.79	7 ms	[n/a]	80
172.16.1.80	5 ms	[n/a]	80
172.16.1.81	5 ms	[n/a]	80
172.16.1.82	7 ms	[n/a]	80
172.16.1.83	6 ms	[n/a]	80
172.16.1.84	[n/a]	[n/s]	[n/s]
172.16.1.85	9 ms	[n/a]	80
172.16.1.86	6 ms	[n/a]	80,8080
172.16.1.87	7 ms	[n/a]	80
172.16.1.88	5 ms	[n/a]	80,8080
172.16.1.89	7 ms	[n/a]	80,8080
172.16.1.90	6 ms	[n/a]	80
172.16.1.91	5 ms	[n/a]	80,8080
172.16.1.92	5 ms	[n/a]	80,8080
172.16.1.93	6 ms	[n/a]	80,8080
172.16.1.94	5 ms	[n/a]	80,8080
172.16.1.95	3 ms	[n/a]	8080
172.16.1.96	[n/a]	[n/s]	[n/s]
172.16.1.97	5 ms	[n/a]	8080
172.16.1.98	5 ms	[n/a]	8080
172.16.1.99	5 ms	[n/a]	80,8080
172.16.1.100	6 ms	[n/a]	80,8080
172.16.1.101	20 ms	[n/a]	80,8080
172.16.1.102	64 ms	[n/a]	80,8080

Started 172.16.3.181 Display: All Threads: 100 (max)

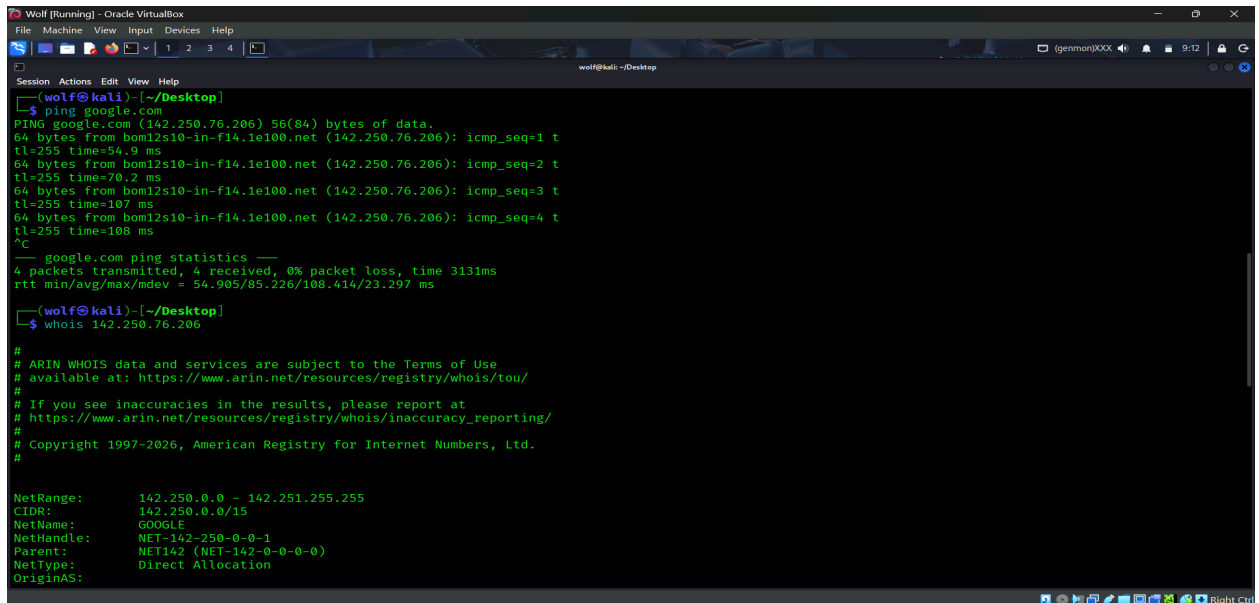
2 . Footprinting (Commands & IP Websites)

- Process of collecting target information

Commands: ping, nslookup, whois, traceroute

Websites: ipinfo.io, geoiptool

- Gives IP, location, ISP, domain details

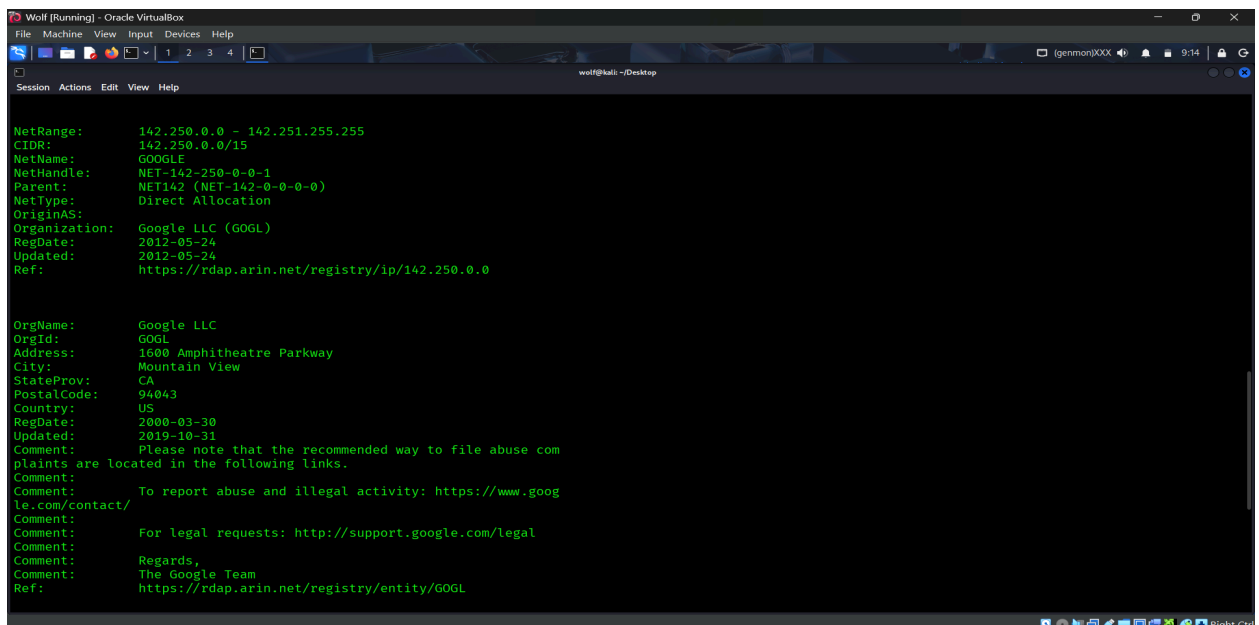


```
Wolf [Running] - Oracle VirtualBox
File Machine View Input Devices Help
wolf@kali: ~/Desktop
$ ping google.com
PING google.com (142.250.76.206) 56(84) bytes of data:
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=1 t
tl=255 time=54.9 ms
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=2 t
tl=255 time=70.2 ms
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=3 t
tl=255 time=107 ms
64 bytes from bom12s10-in-f14.1e100.net (142.250.76.206): icmp_seq=4 t
tl=255 time=108 ms
^C
  google.com ping statistics ---
  4 packets transmitted, 4 received, 0% packet loss, time 313ms
 rtt min/avg/max/mdev = 54.905/85.226/108.414/23.297 ms

$ whois 142.250.76.206

#
# ARIN WHOIS data and services are subject to the Terms of Use
# available at: https://www.arin.net/resources/registry/whois/tou/
#
# If you see inaccuracies in the results, please report at
# https://www.arin.net/resources/registry/whois/inaccuracy_reporting/
#
# Copyright 1997-2026, American Registry for Internet Numbers, Ltd.
#

NetRange:      142.250.0.0 - 142.251.255.255
CIDR:          142.250.0.0/15
NetName:       GOOGLE
NetHandle:     NET-142-250-0-0-1
Parent:        NET142 (NET-142-0-0-0)
NetType:       Direct Allocation
OriginAS:
```



```
Organization:  Google LLC (GOGL)
RegDate:      2012-05-24
Updated:      2012-05-24
Ref:          https://rdap.arin.net/registry/ip/142.250.0.0

OrgName:      Google LLC
OrgId:        GOGL
Address:      1600 Amphitheatre Parkway
City:         Mountain View
StateProv:    CA
PostalCode:   94043
Country:      US
RegDate:      2000-03-30
Updated:      2019-10-31
Comment:      Please note that the recommended way to file abuse com
plaints are located in the following links.
Comment:      To report abuse and illegal activity: https://www.goog
le.com/contact/
Comment:      For legal requests: http://support.google.com/legal
Comment:      Regards,
               The Google Team
Ref:          https://rdap.arin.net/registry/entity/GOGL
```

```
Wolf [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
wolf@kali: ~/Desktop
Session Actions Edit View Help
(wolf@kali)~[~/Desktop]
$ whois google.com
Domain Name: GOOGLE.COM
Registry Domain ID: 2138514_DOMAIN_COM-VRSN
Registrar WHOIS Server: whois.markmonitor.com
Registrar URL: http://www.markmonitor.com
Updated Date: 2019-09-09T15:39:04Z
Creation Date: 1997-09-15T04:00:00Z
Registry Expiry Date: 2028-09-14T04:00:00Z
Registrar: MarkMonitor Inc.
Registrar IANA ID: 292
Registrar Abuse Contact Email: abusecomplaints@markmonitor.com
Registrar Abuse Contact Phone: +1.2086851750
Domain Status: clientDeleteProhibited https://icann.org/epp#clientDeleteProhibited
Domain Status: clientTransferProhibited https://icann.org/epp#clientTransferProhibited
Domain Status: clientUpdateProhibited https://icann.org/epp#clientUpdateProhibited
Domain Status: serverDeleteProhibited https://icann.org/epp#serverDeleteProhibited
Domain Status: serverTransferProhibited https://icann.org/epp#serverTransferProhibited
Domain Status: serverUpdateProhibited https://icann.org/epp#serverUpdateProhibited
Name Server: NS1.GOOGLE.COM
Name Server: NS2.GOOGLE.COM
Name Server: NS3.GOOGLE.COM
Name Server: NS4.GOOGLE.COM
DNSSEC: unsigned
URL of the ICANN Whois Inaccuracy Complaint Form: https://www.icann.org/wicf/
>>> Last update of whois database: 2026-04-25T03:57:01Z <<<

For more information on Whois status codes, please visit https://icann.org/epp

NOTICE: The expiration date displayed in this record is the date the
registrar's sponsorship of the domain name registration in the registry is
currently set to expire. This date does not necessarily reflect the expiration
date of the domain name registrant's agreement with the sponsoring
registrar. Users may consult the sponsoring registrar's Whois database to
view the registrar's reported date of expiration for this registration.

TERMS OF USE: You are not authorized to access or query our Whois
```

```
Wolf [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
wolf@kali: ~/Desktop
Session Actions Edit View Help
(wolf@kali)~[~/Desktop]
$ nslookup google.com
Server: 10.0.2.3
Address: 10.0.2.3#53

Non-authoritative answer:
Name: google.com
Address: 142.250.76.174
Name: google.com
Address: 2404:6800:4009:81a::200e

(wolf@kali)~[~/Desktop]
$ traceroute google.com
traceroute to google.com (142.250.76.174), 30 hops max, 60 byte packets
 1 10.0.2.2 (10.0.2.2) 1.628 ms 1.503 ms 1.444 ms
 2 * * *
 3 * * *
 4 * * *
 5 * * *
 6 * * *
 7 * * *
 8 * * *
 9 * * *
10 * * *
11 * * *
12 * * *
13 * * *
14 * * *
15 * * *
16 * * *
17 * * *
18 * * *
19 * * *
20 * * *
21 * * *
```

3. Reconnaissance via Recon-ng

- Tool for **information gathering (OSINT)**
- Uses modules to collect **domains, emails, subdomains**
- Automates reconnaissance process

```
Wolf [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help
1 2 3 4
wolf@kali: ~/Desktop

Sponsored by...
BLACK HILLS
www.blackhillsinfosec.com

PRACTISEC
www.practisec.com

[recon-ng v5.1.2, Tim Tomes (@lanmaster53)]

[92] Recon modules
[8] Reporting modules
[4] Import modules
[2] Exploitation modules
[2] Discovery modules
[1] Disabled modules

[recon-ng][default] > modules load hackertarget
[recon-ng][default][hackertarget] > info

Name: HackerTarget Lookup
Author: Michael Henriksen (@michenriksen)
Version: 1.1

Description:
```

```
[recon-ng][default][hackertarget] > options set source google.com
SOURCE => google.com
[recon-ng][default][hackertarget] > run

GOOGLE.COM

[*] Country: None
[*] Host: 1.google.com
[*] Ip_Address: 64.233.176.139
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*]
[*] Country: None
[*] Host: 216-239-33-25.google.com
[*] Ip_Address: 216.239.33.25
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*]
[*] Country: None
[*] Host: 216-239-45-10.google.com
[*] Ip_Address: 216.239.45.10
[*] Latitude: None
[*] Longitude: None
[*] Notes: None
[*] Region: None
[*]
[*] Country: None
[*] Host: 216-239-45-32.google.com
[*] Ip_Address: 216.239.45.32
[*] Latitude: None

SUMMARY

[*] 50 total (50 new) hosts found.
[recon-ng][default][hackertarget] > █
```

4 . Vulnerability Finding with Nmap

- Tool for **network scanning & security analysis**
- Finds **open ports, services, vulnerabilities**
- Uses scripts (`--script vuln`) for detection

```
(wolf@kali)-[~/Desktop]
$ nmap -p 22,23,53,139,443,445,2049,8080 103.133.215.103
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 10:29 +0530
Nmap scan report for sindhu1.ewebguru.net (103.133.215.103)
Host is up (0.025s latency).

PORT      STATE      SERVICE
22/tcp    filtered  ssh
23/tcp    closed    telnet
53/tcp    open      domain
139/tcp   filtered  netbios-ssn
443/tcp   open      https
445/tcp   filtered  microsoft-ds
2049/tcp  filtered  nfs
8080/tcp  filtered  http-proxy

Nmap done: 1 IP address (1 host up) scanned in 4.85 seconds
```

-

```

(wolf@kali)~[/Desktop]
$ nmap google.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 09:48 +0530
Nmap scan report for google.com (142.250.76.206)
Host is up (0.0098s latency).
Other addresses for google.com (not scanned): 2404:6800:4009:81b::200e
rDNS record for 142.250.76.206: bom12s10-in-f14.1e100.net
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE
21/tcp    open  ftp
80/tcp    open  http
443/tcp   open  https

Nmap done: 1 IP address (1 host up) scanned in 9.96 seconds

(wolf@kali)~[/Desktop]
$ nmap -p 80 google.com
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 09:49 +0530
Nmap scan report for google.com (142.250.76.206)
Host is up (0.033s latency).
Other addresses for google.com (not scanned): 2404:6800:4009:81b::200e
rDNS record for 142.250.76.206: bom12s10-in-f14.1e100.net

PORT      STATE SERVICE
80/tcp    open  http

Nmap done: 1 IP address (1 host up) scanned in 4.76 seconds

```

```

(wolf@kali)~[/Desktop]
$ nmap -sV 142.250.76.206
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 09:50 +0530
Nmap scan report for bom12s10-in-f14.1e100.net (142.250.76.206)
Host is up (0.014s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
21/tcp    open  ftp?
80/tcp    open  http         gws
443/tcp   open  ssl/https    gws
2 services unrecognized despite returning data. If you know the service/version, please submit the following fingerprints at https://nmap.org/cgi-bin/submit.cgi?new-service :
=====NEXT SERVICE FINGERPRINT (SUBMIT INDIVIDUALLY)=====
SF-Port80-TCP:V=7.98%I=7%D=4/26%Time=69ED929E%P=x86_64-pc-linux-gnu%r(GetR
SF:request,164E,"HTTP/1.0\x20200\x20OK\r\nDate:\x20Sun,\x2026\x20Apr\x2020
SF:26\x204:20:48\x20GMT\r\nExpires:\x20-1\r\nCache-Control:\x20private,\x
SF:20max-age=0\r\nContent-Type:\x20text/html;\x20charset=ISO-8859-1\r\nCon
SF:tent-Security-Policy-Report-Only:\x20object-src\x20'none';base-uri\x20'
SF:self';script-src\x20'nonce-1x0TE764eNan8cXcmcUa1w'\x20strict-dynamic \
SF:\x20'report-sample'\x20'unsafe-eval'\x20'unsafe-inline'\x20https:\x20htt
SF:p;;report-uri\x20https://csp.withgoogle.com/csp/gws/other-hp\r\nP3P:\
SF:\x20CP="This\x20is\x20not\x20a\x20P3P\x20policy!\x20See\x20g.co/p3phel
SF:p\x20for\x20more\x20info\","*\r\nServer:\x20gws\r\nX-XSS-Protection:\x20
SF:0\r\nX-Frame-Options:\x20SAMEORIGIN\r\nSet-Cookie:\x20_secure-STRP=AEE
SF:P7glcug10nLrsPMU3mh_R4E5FpBelsgNAS4quPS2feyBtFy13WnltTj14Rg-bPm0ADqkyps
SF:GceNz4Db0_PlGg-6dMu0PDDw;\x20expires=Sun,\x2026-Apr-2026\x204:25:48\x2
SF:0GMT;\x20path=/;\x20domain=.google.com;\x20Secure;\x20SameSite=strict

```



```
Wolf [Running] - Oracle VirtualBox
File Machine View Input Devices Help
1 2 3 4
wolf@kali: ~/Desktop

Nmap done: 1 IP address (1 host up) scanned in 191.67 seconds

(wolf@kali) [~/Desktop]
$ nmap -A 142.250.76.206
Starting Nmap 7.98 ( https://nmap.org ) at 2026-04-26 09:57 +0530
Nmap scan report for bom12s10-in-f14.1e100.net (142.250.76.206)
Host is up (0.0069s latency).
Not shown: 997 filtered tcp ports (no-response)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp?
80/tcp    open  http    gws
_http-server-header: gws
_fingerprint-strings:
  GetRequest:
    HTTP/1.0 200 OK
    Date: Sun, 26 Apr 2026 04:27:27 GMT
    Expires: -1
    Cache-Control: private, max-age=0
    Content-Type: text/html; charset=ISO-8859-1
    Content-Security-Policy-Report-Only: object-src 'none';base-uri 'self';script-src 'nonce-nq3Nu4kfHKyBQOMAIwnoAw' 'strict-dynamic' 'report-sample' 'unsafe-eval' 'unsafe-inline' https: http:;report-uri https://csp.withgoogle.com/csp/gws/other-hp
    P3P: CP="This is not a P3P policy! See g.co/p3phelp for more info."
    Server: gws
    X-XSS-Protection: 0
    X-Frame-Options: SAMEORIGIN
    Set-Cookie: __Secure-STRP=AEEP7gLNyWPV6qH_jlZUSicZw405poRNlqWfCTnBpoUA0eXi3iuuNyZmiExqVGe4qEoq849nz2_UHhwEfahB2JafT5MHeZgHeA; expires=Sun, 26-Apr-2026 04:32:27 GMT; path=/; domain=.google.com; Secure; SameSite=strict
    Set-Cookie: AEC=AaJma5sYPbBg6aK91o01h7p5BrckY6hj7nXN31mQrfSH3m0JjDaxJiXcaCM; expires=Fri, 23-Oct-2026 04:27:27 GMT; path=/; domain=.google.com; Secure;
  HttpOn
    HTTPOptions:
      HTTP/1.0 405 Method Not Allowed
      Content-Type: text/html; charset=UTF-8
      Referrer-Policy: no-referrer
      Content-Length: 1592
      Date: Sun, 26 Apr 2026 04:27:27 GMT
      <!DOCTYPE html>
      <html lang=en>

SF:expires=Fri,\x2023-Oct-2026\x2004:27:33\x20GMT;\x20");
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Device type: bridge|VoIP adapter|general purpose
Running (JUST GUESSING): Oracle Virtualbox (98%), Slirp (98%), AT&T embedded (95%), QEMU (94%)
OS CPE: cpe:/o:oracle:virtualbox cpe:/a:danny_gasparovski:slirp cpe:/a:qemu:qemu
Aggressive OS guesses: Oracle Virtualbox Slirp NAT bridge (98%), AT&T BGW210 voice gateway (95%), QEMU user mode network gateway (94%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

TRACEROUTE (using port 80/tcp)
HOP RTT ADDRESS
1 1.75 ms bom12s10-in-f14.1e100.net (142.250.76.206)

OS and Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 317.94 seconds
```

3. What is Proxy Server & V.P.N. With Detailed Explanation.

Proxy Server & VPN (Point-wise)

◆ Proxy Server

- Acts as an **intermediary** between user and internet
- Request flow: **User → Proxy → Website**
- Hides **user's IP address**
- Used for:
 - Anonymous browsing
 - Bypassing restrictions
 - Web testing (Burp Suite)
- **No encryption** (in most cases)
- **Less secure**

VPN (Virtual Private Network)

- Creates a **secure encrypted tunnel**
- Connection: **User → VPN → Internet**
- Hides IP and **encrypts data**
- Used for:
 - Secure browsing
 - Privacy protection
 - Public WiFi safety
- ISP cannot see user activity
- **Highly secure**

Key Differences

- Proxy:
 - No encryption
 - Lower security
 - Faster
 - App-level use
- VPN:
 - Encryption present
 - High security
 - Slightly slower
 - Works system-wide

Conclusion

- **Proxy** → Basic anonymity
- **VPN** → Security + Privacy
- VPN is **more secure than proxy**

4. What are the Different Types of Malwares. Explain it with detail.

Types of Malware (Malicious Software)

Malware is any software designed to **harm, exploit, or gain unauthorized access** to a computer system.

Creeper Virus

- Considered the first computer virus (1971)
- Created by Bob Thomas
- Infected ARPANET systems (early internet)

1. Virus

- Attaches itself to **files or programs**
- Spreads when infected file is executed
- Can **corrupt or delete data**
- Example: File-infecting virus

2. Worm

- Self-replicating malware
- Spreads **automatically over networks**
- Does not need user action
- Slows down systems and networks

3. Trojan Horse

- Disguised as **legitimate software**
- Tricks users into installing it
- Creates **backdoor access** for attackers
- Example: Fake apps, cracked software

4. Ransomware

- Encrypts user data
- Demands **payment (ransom)** to unlock files
- Very dangerous for organizations
- Example: WannaCry

5. Spyware

- Secretly monitors user activity
- Collects sensitive data (passwords, browsing habits)
- Sends data to attacker

6. Adware

- Displays unwanted advertisements
- Tracks user behavior for marketing
- Slows down system performance

7. Keylogger

- Records keystrokes (keyboard input)
- Steals passwords, credit card details
- Works silently in background

Conclusion

- Malware comes in many forms with different purposes
- Main goals: **data theft, damage, unauthorized access**
- Prevention: **antivirus, updates, safe browsing**

5. What is TOR Network and Practically Explain how to Create Identities.

TOR Network (The Onion Router)

Definition

- TOR is a privacy-focused network that allows users to browse the internet anonymously
- It hides the user's identity by routing traffic through multiple servers (nodes)

How TOR Works

- Uses onion routing (multi-layer encryption)
- Data passes through 3 main nodes:
 1. Entry Node – knows your IP
 2. Relay Node – passes data
 3. Exit Node – sends request to internet

Features of TOR

- Provides **anonymity & privacy**
- Bypasses censorship
- Access to **.onion (dark web) sites**
- Free and open-source

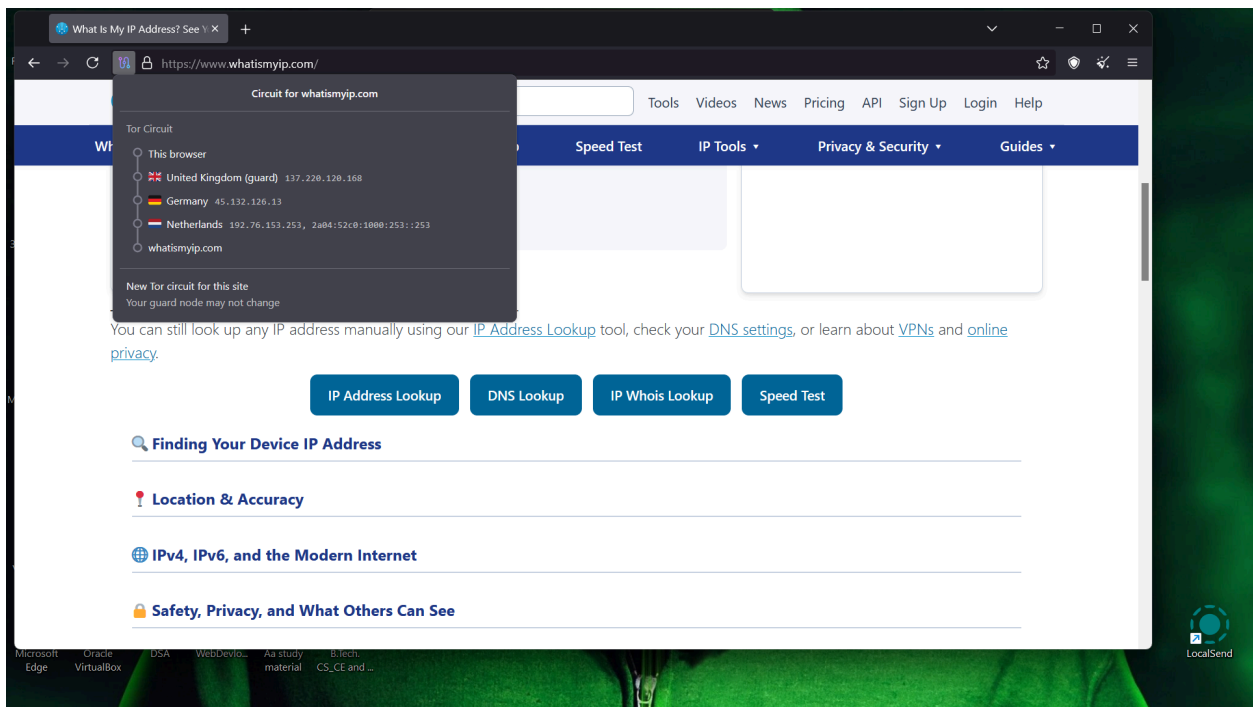
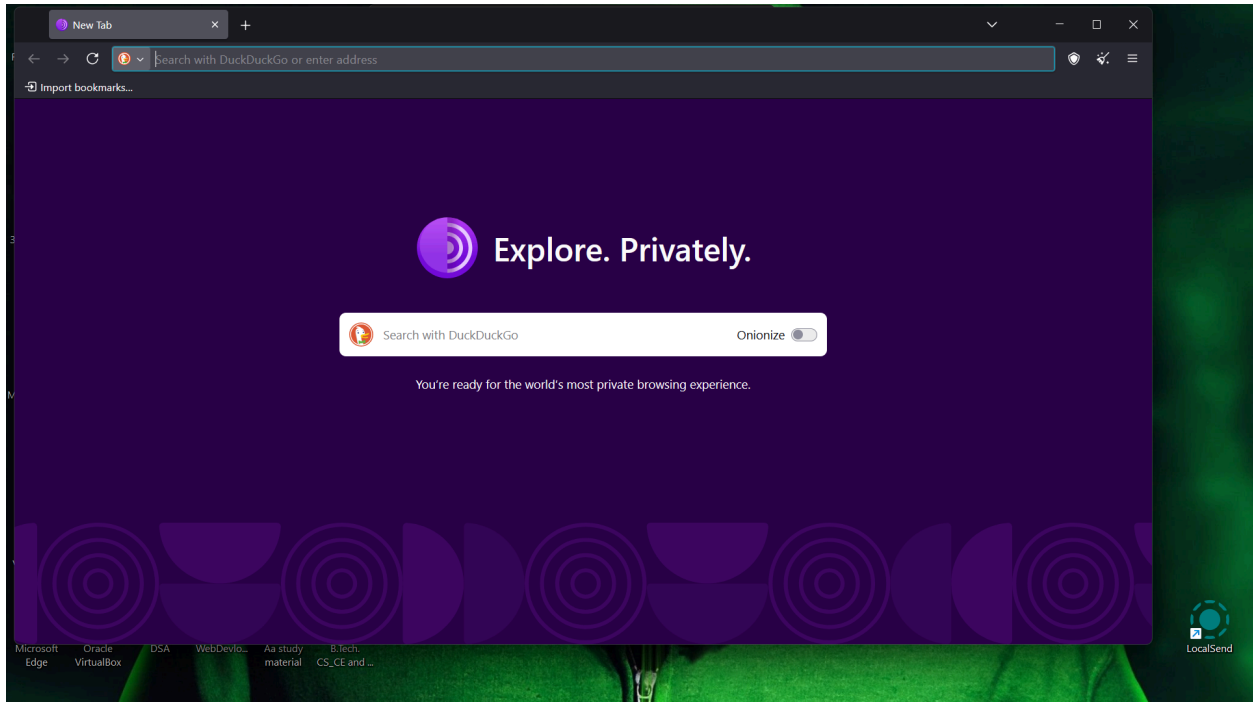
Uses

- Secure communication
- Research and OSINT
- Bypassing restrictions
- Protecting identity (journalists, researchers)

“Creating Identities” in TOR

Important:

- . This does **NOT** mean creating fake illegal identities.
- . It means maintaining **anonymous digital presence**.



Key Idea

- TOR helps create **anonymous sessions**, not fake people
- Identity = **online pseudonym + hidden IP**

Conclusion

- TOR is a powerful tool for **privacy and anonymity**
- It uses **multi-layer encryption** to hide user identity
- “Creating identities” means **maintaining anonymous profiles safely**, not illegal impersonation

6. Explain the Password Attacks.

Password Attacks

Password attacks are techniques used by attackers to **steal, guess, or crack user passwords** to gain unauthorized access to systems.

1. Brute Force Attack

- Tries **all possible password combinations**
- Uses automated tools
- Time-consuming but effective

2. Dictionary Attack

- Uses a **predefined list of common passwords**
- Faster than brute force
- Example: “123456”, “password”

3. Phishing Attack

- Tricks users into giving passwords
- Uses **fake emails or websites**
- Example: Fake login page

4. Keylogger Attack

- Records keyboard input
- Captures passwords secretly
- Runs in background

5. Shoulder Surfing

- Observing someone typing password
- Done in public places

6. Credential Stuffing

- Uses leaked usernames/passwords
- Tries them on multiple websites
- Works because people reuse passwords

7. Rainbow Table Attack

- Uses precomputed hash tables
- Cracks encrypted passwords quickly

Prevention Methods

- Use **strong passwords** (mix of letters, numbers, symbols)
- Enable **2-Factor Authentication (2FA)**
- Avoid password reuse
- Use **password managers**
- Beware of phishing links

Conclusion

Password attacks are common methods to break security. Strong passwords and security practices help **prevent unauthorized access**.

7. What is Dos & DDos Attack.

DoS & DDoS Attack

1. DoS (Denial of Service) Attack

- A **DoS attack** is an attempt to make a system or website **unavailable** to users
- Done by **flooding the server** with too many requests
- Comes from **a single system (attacker)**

Result:

- Server becomes slow or crashes
- Legitimate users cannot access the service

2. DDoS (Distributed Denial of Service) Attack

- A **DDoS attack** is similar to DoS but more powerful
- Uses **multiple systems (botnet)** to attack a target
- Traffic comes from many sources at the same time

Result:

- Server is overloaded quickly
- Very difficult to stop

How They Work

- Attackers send a huge number of requests
- Server resources (CPU, bandwidth) get exhausted
- System becomes unavailable

Key Differences

Feature	DoS Attack	DDoS Attack
Source	Single system	Multiple systems
Power	Less	Very high
Detection	Easier	Difficult
Example	One attacker	Botnet (many devices)

Prevention Methods

- Use **firewalls & IDS/IPS**
- Rate limiting
- Load balancing
- Use CDN (Content Delivery Network)
- Monitor unusual traffic

Conclusion

- DoS and DDoS attacks aim to **disrupt services**
- DDoS is more dangerous due to **multiple attacking sources**

8. What is sniffing and their types.

Sniffing and Its Types

What is Sniffing?

- Sniffing is the process of **capturing and monitoring network traffic (data packets)**
- It is used to **analyze data flowing in a network**
- Can be **legitimate (for security)** or **malicious (for stealing data)**

How Sniffing Works

- Data travels in the form of **packets** over a network
- A sniffer tool captures these packets
- Attacker or admin can **read sensitive information**
 - Passwords
 - Emails
 - Login data

Types of Sniffing

1. Active Sniffing

- Used in **switched networks**
- Attacker interacts with network to capture data
- Techniques used:
 - ARP spoofing
 - MAC flooding
- More complex

2. Passive Sniffing

- Used in **hub-based networks**
- Attacker just listens to traffic
- No interaction needed
- Easier to perform

Prevention Methods

- Use **encryption (HTTPS, VPN)**
- Use secure protocols (SSH instead of Telnet)
- Network security tools (IDS/IPS)
- Avoid public WiFi

Conclusion

- Sniffing is a technique to **capture network traffic**

- It can be **useful for security** or **dangerous if misused**
- Proper security measures can prevent data theft

9. Explain about Kali Linux, Metasploit framework and its usage.

Kali Linux

Definition

- Kali Linux is a **Debian-based Linux distribution** used for **penetration testing and security auditing**
- Developed by **Offensive Security**

Features

- Comes with **600+ security tools** pre-installed
- Open-source and free
- Supports **wireless attacks, web testing, forensics**
- Regular updates

Common Tools in Kali

- Nmap → Network scanning
- Wireshark → Packet analysis
- Burp Suite → Web testing
- Metasploit → Exploitation

Uses

- Ethical hacking
- Vulnerability assessment
- Digital forensics
- Security research

Metasploit Framework

Definition

- Metasploit is a **powerful penetration testing framework**
- Used to **find, exploit, and validate vulnerabilities**

Components

- **Exploit** → Code to take advantage of vulnerability
- **Payload** → Code executed after exploitation
- **Auxiliary** → Scanning and other functions
- **Encoder** → Evades detection

Types of Payloads

- **Singles** → Independent payload
- **Staged** → Sent in parts (more flexible)

Uses

- Exploiting system vulnerabilities
- Testing security of networks
- Gaining access (for ethical testing)
- Post-exploitation activities

Conclusion

- **Kali Linux** is a complete platform for security testing
- **Metasploit** is one of its most powerful tools for exploitation
- Both are widely used in **ethical hacking and cybersecurity**

CONCLUSION

This minor project provided practical understanding of cybersecurity concepts, tools, and techniques.

It helped in learning how attackers exploit vulnerabilities and how organizations can protect systems.

All activities were performed in a **controlled lab environment for educational purposes only**.

REFERENCES

- Kali Linux Official Documentation
- OWASP Top 10 Security Guide
- Nmap Documentation
- Metasploit Framework Guide
- Cyber Security Academic Materials